MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 249C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, mercury-vapor

See figure 1

Mounting position: Base down

Weight: 3-ounces (85 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	THg	tk	tave	F
Unit:	Vac	v	a	Adc	°C	sec	sec (see note 1)	Hz
Maximum:	2.625	7,500	2.5	0.64	70	---	5	150
Minimum:	2.375	---	---	---	20	15	---	---
TEST CONDITIONS:	2.5	---	---	---	40 to 60	30	---	60

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	6.75	8.25	Aac
1231	Peak emission by voltage drop	-	ib = 4a	0.65	II	etd	---	18	v
1353	Operation of rectifiers (1)	2	epx = 8.5 kv; Ib = 0.64 Adc; t = 60 sec (min)	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Vibration	-	No voltages applied	---	---	---	---	---	---
1353	Operation of rectifiers (2)	2	epx = 8.5 kv; Ib = 0.64 Adc; THg = 68° to 72°C; t = 300 sec (min)	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group D; epx = 7.5 kv; Ib = 0.64 Adc; ib = 2.5 a; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission by voltage drop	-		---	---	etd	---	22	v

NOTES:

1. tave: Maximum averaging time for anode current.
2. Tube shall operate without signs of arc-back or flashing after the first half minute of test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

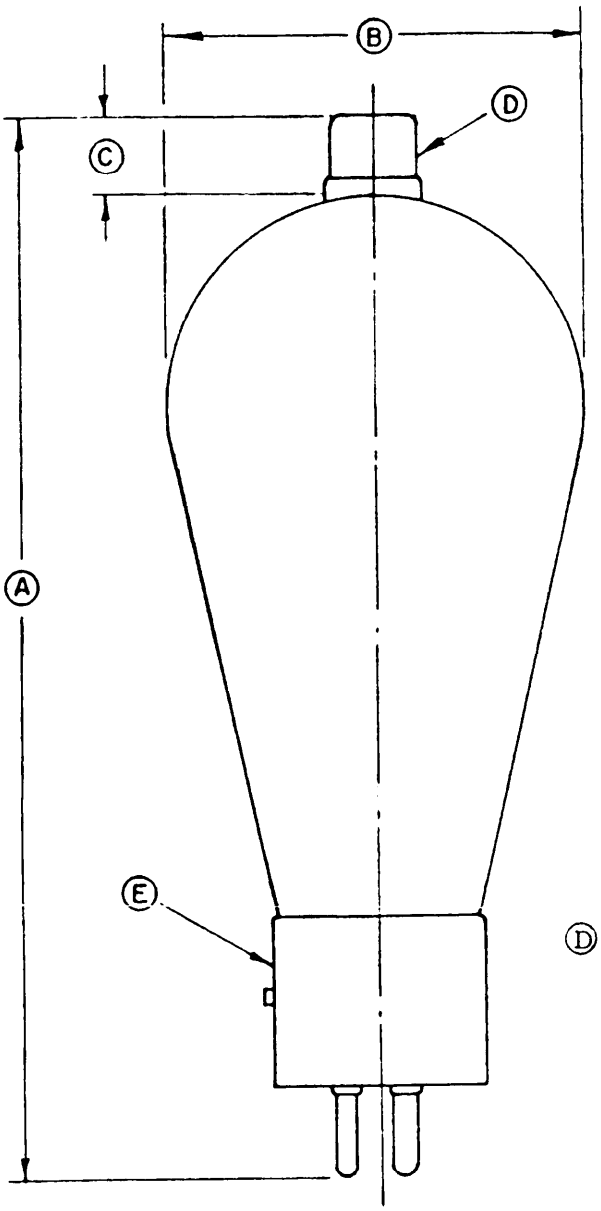
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2929)

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PIN CONNECTIONS	
Pin No.	Element
1 & 2	f
3 & 4	f
cap	a

	Dimensions in inches with metric equivalents (mm) in parentheses	
Ltr	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.813 (173.05)	7.625 (193.68)
B		2.688 (68.28)
Quality conformance inspection, part 3		
C	.969 nom (24.61 nom)	
D	Cap: C1-5 (EIA)	
E	Base: A4-10 (EIA)	

NOTE:
These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1; for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 249C.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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